Status	Finished
Started	Thursday, 29 October 2026, 9:42 PM
Completed	Thursday, 29 October 2026, 9:45 PM
Duration	3 mins 15 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Backtracking
b. Dynamic programming
c. Divide and conquer
d. Greedy method

The correct answer is:
Divide and conquer

Question **2**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are ignored
b. Each part is searched for a specific element
c. Each part is sorted independently
d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question **3**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Arranging data in a particular format
- b. Inserting data into a list
- c. Searching for data in a list
- d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **4**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Searching
- b. Selection
- c. Duplicates
- d. All of the above

The correct answer is: All of the above

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It builds a sorted array one element at a time
- b. It compares adjacent elements and swaps them if necessary
- c. It divides the list into two halves, sorts each half, and then merges them
- d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **6**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are split again into smaller sublists
b. They are compared element by element
c. They are combined to form a single sorted list
d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **7**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Sorting
b. Searching
c. Merging
d. Complexity

The correct answer is:
Complexity

Question **8**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It allows for efficient parallel processing and sorting of data
b. It simplifies the sorting process by using only one pass
c. It only works on small datasets
d. It avoids the need for recursion

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **9**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is difficult to implement
b. Bubble Sort cannot handle duplicate elements
c. Bubble Sort does not guarantee sorted order
d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **10**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Bubbling up the largest element to its correct position with each pass
b. Using the divide-and-conquer approach
c. Sorting data in a non-sequential manner
d. Being the most efficient sorting algorithm

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **11**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Quick Sort
b. Merge Sort
c. Linear Search
d. Bubble Sort

The correct answer is: Merge Sort

Question **12**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By breaking the input into smaller parts and merging them
- b. By using the bubble-up method
- c. By sorting data in a single pass
- d. By comparing elements sequentially

The correct answer is: By breaking the input into smaller parts and merging them

Question **13**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Distribution
- b. Insertion
- c. Selection
- d. Extraction

The correct answer is:
Insertion

Question **14**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. order()
- b. arrange()
- c. sort()
- d. sorted()

The correct answer is: sorted()

Question **15**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Dividing the input into smaller subproblems
c. Comparing each element with the others
d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems